

V22: the implemented multiscale correction

Verbatim source excerpt: reconstruction_v22/operator.py, lines 119–138.

```

anchor = quadratic[control_k]
normals = anchor["normal"]
delta_vectors = np.stack([quadratic[k]["correction"][:, None] * quadratic[k]["normal"]
                           for k in actual_ks], axis=1)
projected = np.einsum("nsi,ni->ns", delta_vectors, normals)
point_variance = np.median(np.stack([quadratic[k]["noise_variance"] for k in actual_ks], axis=1), axis=1)
prediction_variances = np.stack([quadratic[k]["prediction_variance"] for k in actual_ks], axis=1)
# Numerical floor is dimensional and tiny; it is not a sensor-noise guess.
floor = np.maximum(anchor["tangent_scale"]**2 * 1e-24, np.finfo(float).tiny)
safe_variances = np.maximum(prediction_variances, floor[:, None])
# Equivalent inverse-variance ratios, without overflow for identical points.
precision = safe_variances.min(axis=1, keepdims=True) / safe_variances
weights = precision / precision.sum(axis=1, keepdims=True)
mean_delta = np.sum(weights * projected, axis=1)
mean_vector = mean_delta[:, None] * normals
disagreement = np.sum(weights * np.sum((delta_vectors - mean_vector[:, None])**2, axis=2), axis=1)
# Fits share observations; do not claim a 1/number_of_scales variance gain.
prediction_variance = np.sum(weights * prediction_variances, axis=1)
alpha = point_variance / (point_variance + prediction_variance + disagreement + floor)
consensus = points + (alpha * mean_delta)[:, None] * normals

```

The applied displacement is $\alpha_i d_i \mathbf{n}_{i,64}$, where

$$\alpha_i = \frac{v_{n,i}}{v_{n,i} + v_{p,i} + v_{s,i} + \varepsilon_i}.$$

Shared observations are not treated as independent evidence: prediction variance is not divided by the number of scales. The factor α is heuristic damping, not a calibrated probability or a rejection gate.

Published comparison against identity SUMMARY.json / results.confirmation.comparisons

```

{
  "identity": {
    "paired": 24,
    "relative_gain": 0.005980615532293565,
    "win_fraction": 0.8333333333333334,
    "recall_difference": -0.003325902214863378,
    "passed": false
  }
}

```

passed: false means the predeclared composite improvement criterion was not met; it does not mean execution failed. Gains and recall differences in this JSON are fractions, not percentages.